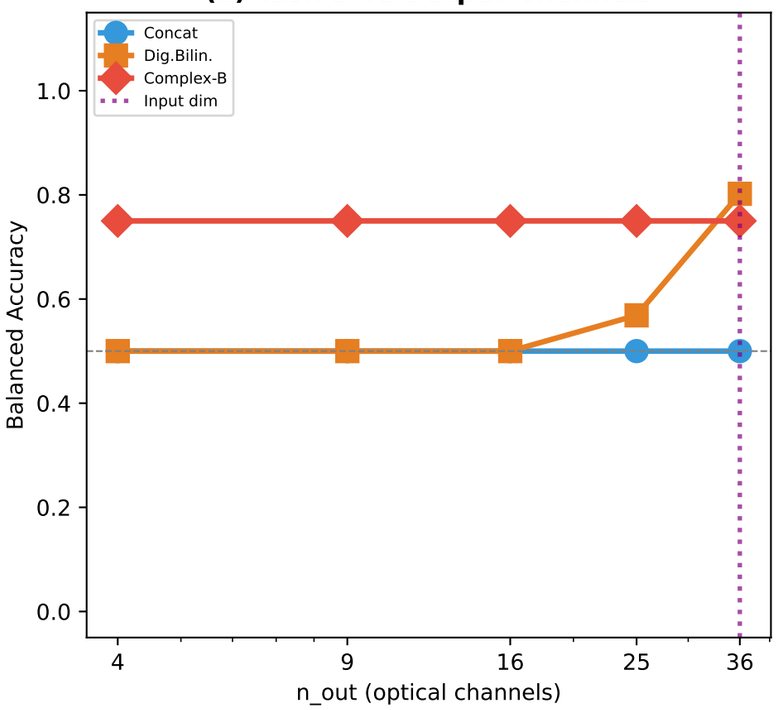
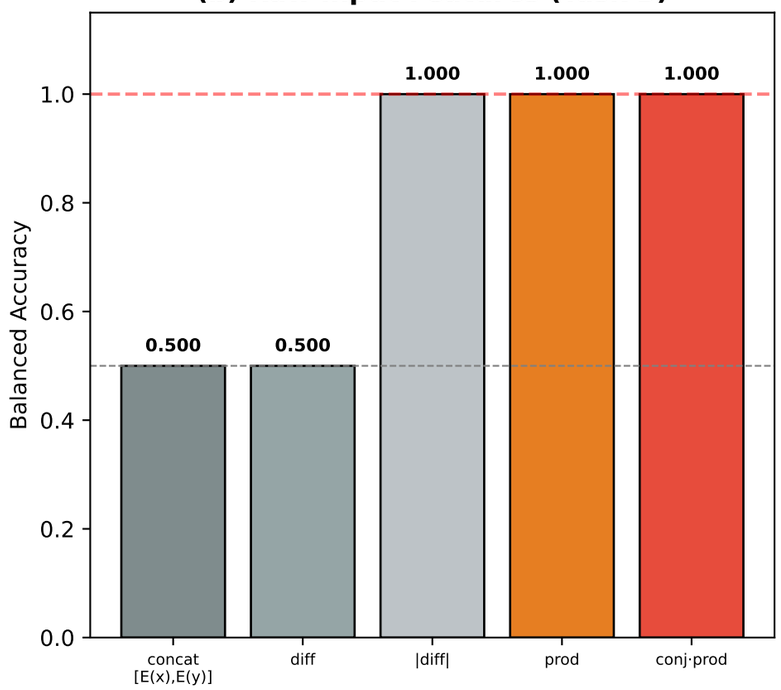
Task-Regime Summary for Optical Random Projection Features

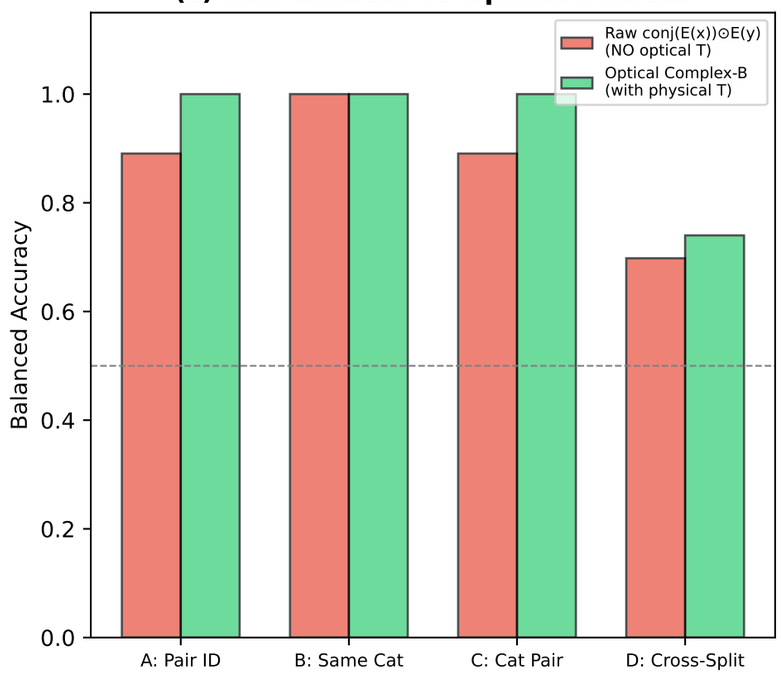
(a) Task B vs Output Channels



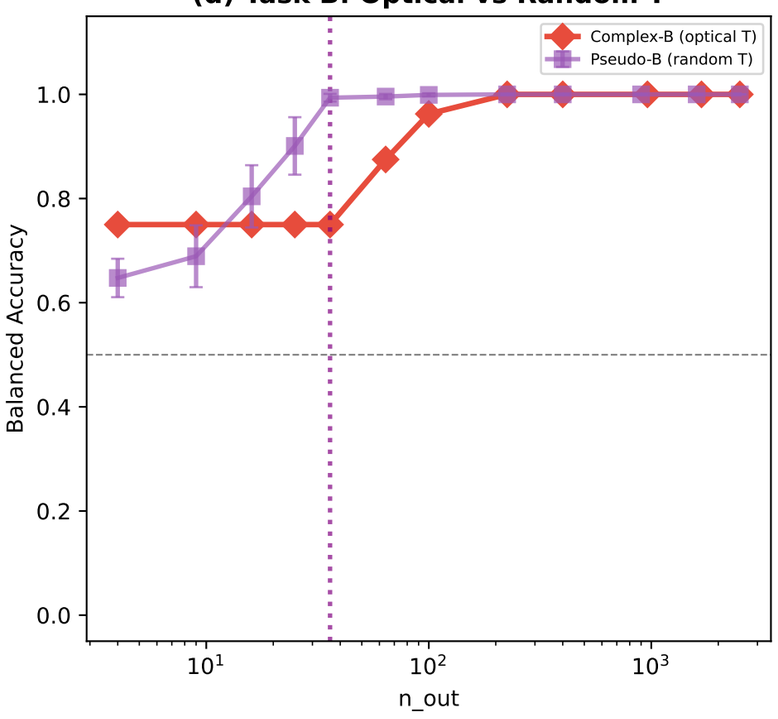
(b) Raw Input Features (Task B)



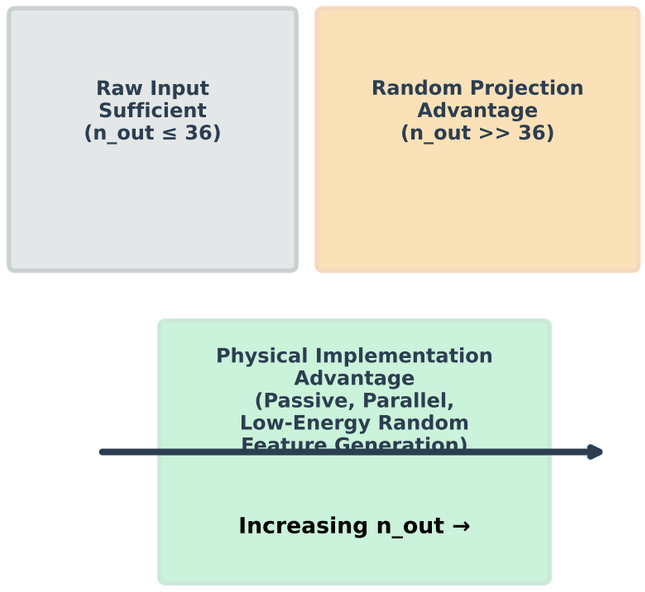
(c) Raw Bilinear vs Optical Bilinear



(d) Task B: Optical vs Random T



(e) Advantage Boundary Map



(f) Key Findings

1. Raw input features (no T) do not separate Task B in this baseline (same-category acc ≈ 0.50)

2. At n_out ≤ 36 (≤ input dim), Complex-B still outperforms raw inputs on Task B → Low-dim optical readout adds information in this tested setting

3. Both physical T and random T achieve ~1.0 at sufficient n_out — the advantage is in random-projection feature family, not uniquely the measured T matrix

4. Physical implementation relevance: passive, parallel, fixed optical features without a digital matrix multiply during optical propagation